

Exercise set 3, Statistical Methods in Physics, 2025

Assessment criteria: for “passed”, solutions to all exercises marked “mandatory” must be handed in before **10 am, January 14th, 2026**. In addition, the student must choose and solve the “advanced” exercise from this set or advanced exercises from previous sets. For “passed with distinction”, solutions to all exercises marked “mandatory” must be solved before **10 am, January 14th** and in addition, the student must choose two out of three “advanced” exercises from this and the coming exercise sets.

All exercises must be correct (after iterating with the teacher) on **January 30th**.

1. Systematic uncertainties: A PhD student measures the height of a table-top Xmas tree with a steel ruler and finds it to be 99.6 cm. The space between the markers of the ruler is 1 mm. The steel ruler is calibrated at room temperature (20° C). However, the student’s office is at Ångström, which is considerably colder (16°) since they have to work with the window open due to bad ventilation. The student looks up the expansion coefficient of steel in a table and finds it to be $0.0005 \pm 0.0001 \text{ } ^\circ\text{C}^{-1}$.

- a) **(Mandatory)** What is the height of the tree if the student corrects for the thermal expansion of the steel ruler?
- b) **(Mandatory)** Calculate the total systematic uncertainty based on the sources which have been mentioned directly or indirectly.
- c) **(Mandatory)** Can you think of some other source of systematic uncertainty that we cannot quantify with the given information?
- d) **(Mandatory)** So far, we have focused on the precision of the Xmas tree height. Describe a situation where we would be interested in the tolerance, rather than the precision?

2. Monte Carlo methods: In this exercise, you are supposed to generate a Gaussian distribution, fold it with a response function and then unfold it using two different methods.

- a) **(Mandatory)** The **polar method** is a simple and efficient procedure to transform a uniform random distribution into a Gaussian distributed one. The algorithm works as follows:

1. Generate two random numbers $u, v \in (-1, 1)$ Calculate $r = u^2 + v^2$
2. If $r > 1$ or $r = 0$: Go back to step 1.
3. Calculate $m = \sqrt{\frac{-2 \ln r}{r}}$.
4. Calculate $x_1 = u \cdot m$ and $x_2 = v \cdot m$.

x_1 and x_2 are two independent, normally distributed random numbers, *i.e.* the expectation value is 0.0 and the standard deviation is 1.0. The algorithm always produces two random numbers per application, not just one. It is also straightforward to use this method to generate a general Gaussian with a given μ and σ for example by defining $x_1 = \mu + \sigma \cdot u \cdot m$.

Implement a function to transform a uniform distribution into a Gaussian with mean $\mu = 5.0$ with a width $\sigma = 2.0$. Plot distributions for u, v as well as for x_1, x_2 . Check that the generated distribution has the desired mean and width, using for example a simple LSQ fit. You may use pre-written software for fitting procedure.

More exercises on the next page

3. Unfolding In an experiment, a distribution $f(x)$ of a physical quantity x is measured with a detector whose response (“transfer”) function is $t(x'; x)$. The variables x and x' can be organised into histograms, with bins $1, \dots, m$ and $1, 2, \dots, n$, respectively. For this binned case the function t becomes a matrix R of elements R_{ij} where $i = 1, \dots, n$ and $j = 1, \dots, m$.

R is given below and parameterizes both the smearing due to limited detector resolution and the event loss due to limited efficiency. The smearing implies means the probability to observe the true value x for a measured value x' in a certain bin is 40%, except for the first and n :th value of x where it is 65%. There is also an overall efficiency of 90% over the full x range.

$$R = 0.9 \cdot \begin{pmatrix} 0.65 & 0.25 & 0.1 & 0 & \dots & 0 & \dots & 0 \\ 0.25 & 0.4 & 0.25 & 0.1 & 0 & 0 & \dots & 0 \\ 0.1 & 0.2 & 0.4 & 0.2 & 0.1 & 0 & \dots & 0 \\ 0 & 0.1 & 0.2 & 0.4 & 0.2 & 0.1 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & 0.1 \\ \vdots & \dots & 0 & 0.1 & 0.25 & 0.4 & 0.25 & \vdots \\ 0 & \dots & \dots & 0 & 0.1 & 0.25 & 0.65 & 0 \end{pmatrix}$$

Let $m = n = 10$. Take the function you implemented in exercise 2 for the Gaussian of mean $\mu = 5.0$ with a width $\sigma = 2.0$.

- (Mandatory)** Generate a distribution with 1000 events with these properties, organize them in a histogram with 10 bins and “smear” it with the response matrix R : $f'(x') = Rf(x)$. Extract the parameters of the **folded** distribution (mean and width), by *e.g.* a LSQ fit.
- (Mandatory)** Generate 1000 **new** events of the variable x' , using the parameters extracted from the folded histogram in 3a). Note that this distribution should NOT be multiplied by R since it should already have the properties (i.e. mean and width) of a folded distribution. Hence, this new distribution “simulates” a distribution that has been distorted by the detector resolution and efficiency. Now use **correction factor method** to unfold this distribution to obtain an estimate of “the truth”, i.e. $f(x)$. Please also calculate the uncertainties (covariance matrix). Illustrate with histograms/plots on all levels of generating and unfolding.
- (Advanced)** Take the 1000 events that you generated in c) of the variable x' . Now unfold the distribution for the “truth”, including uncertainties, using the **matrix inversion method**. Comment on any discrepancies compared to the correction factor method. Illustrate the solution with histograms/plots on all levels of generating and unfolding.

Guidelines for solutions

- The solutions should be provided in PDF format (not as a notebook) in a private message either on Studium or Slack.
- The solutions should include the step-by-step explanations and motivations.
- If you generate your pdf file from a python notebook, please make sure that you add blocks of text for explanations and analytical mathematical expressions,
- Please include code snippets in your pdf file if relevant for your solution. You do not need to use python but can use ROOT, MATLAB or another language of choice.
- If you want to write the explanations by hand, please provide a clear and clean text.
- Hand-drawn figures are only suitable for explaining concepts and cannot be used for extracting numbers. If you use a graphical method for estimating a quantity, please display the figure from the software in which you extract the numerical solution.
- You are allowed to collaborate, but should hand in your solutions individually and it should be clarified with whom you have collaborated.
- In general, generative AI is **not** allowed for solving exercises, **BUT** so-called code completion is allowed.